

STRUCTURE AND COMPUTATION IN SMALL CHESS POLICIES

Research draft

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ABSTRACT

We compare four structural biases in small chess decision networks: explicit candidate consequences, biological recurrent wiring, exact computation reuse, and action-conditioned relation transport. In a twelve-fit candidate comparison, explicit successor differences miss predeclared engine-quality and paired-game criteria despite sensitivity to action-successor alignment. A separate eighteen-adaptor experiment compares a fixed biological subgraph against three degree-preserving rewires and additional recurrent controls on frozen backbones. The biological topology fails its frozen continuation criterion. A factorized relation network then separates global metadata from local recurrence, allowing exact dependency-cone updates in real arithmetic. Its floating-point implementation reduces logical recurrent-node updates by 60.6–72.0%. A separately frozen batched baseline is numerically equivalent and makes the original incremental path 2.91 times slower in a paired complete-loop comparison. A fifteen-fit comparison of two-endpoint relation transport against routing, overlap and topology controls fails its development teacher-agreement criterion. An eighteen-fit joint-training follow-up also fails its criterion; its advantage over equally fine-tuned direct prediction nearly vanishes on shifted positions. A matched 24-fit pin-factor comparison passes only two of sixteen checks, both against the frozen backbone, and fails its continuation criterion. Every experiment retains its original criterion, unsuccessful arms and costs. Together, these development results distinguish input dependence, structural attribution, logical work, measured latency and chess strength. They do not yet establish a novel architecture advantage. Fresh confirmation and direct comparisons with the closest published methods remain necessary.

1 RESEARCH QUESTION

Chess has an exact transition simulator, a finite legal action set, strong reference engines, and controlled interventions on input information. These properties make it useful for testing architectural mechanisms. They also make weak attribution easy: a legal-action mask can guarantee legal moves, a native simulator can expose future states, and a teacher can supply strong labels without the student learning planning.

We investigate four questions. Does explicitly encoding each candidate’s consequences improve decisions enough to justify its cost? Does biological wiring improve a policy beyond degree-matched random wiring after sharing the backbone, input interface, initialization and optimization? Can a representation be designed so that different candidate consequences reuse exact computations without invalidating the whole network? Does action-conditioned transport through root-board relations improve a move head beyond routing and overlap ablations or degree-preserving rewires?

The experiments form a sequential development program. Each comparison has its own frozen protocol and panel; they are not a shared leaderboard. Results that fail a predeclared continuation rule remain failures. Subsequent experiments cannot retroactively convert those outcomes into successful confirmation. The locality architecture is new code in this project, not yet a demonstrated new contribution relative to the literature.

2 RELATED WORK AND BOUNDARIES

Recurrent computation on chess, including extrapolation from easier to harder problems by increasing inference iterations, is established by Schwarzschild et al. (2021). ChessBench studies amortized chess decisions from large-scale teacher data (Ruoss et al., 2024); matching a teacher move is not the same as winning a game. AlphaGateau studies graph representations, edge features and policy outputs for chess (Rigaux and Kashima, 2024). Neither graph inputs nor tied recurrence alone provides novelty here. Probing and activation interventions have also found evidence of future-move representations in Leela (Jenner et al., 2024); exposing native successors in our inputs does not by itself establish learned look-ahead.

FlyGM instantiates a fly connectome as a graph controller for simulated locomotion (Jin et al., 2026). ChessFly is a prior public artifact using a FlyWire-derived graph (ChessFly, 2026). Our controlled topology experiment asks a narrower causal question than comparing heterogeneous published policies. Our induced descending-neuron subgraph omits its original sensory inputs and uses an artificial board interface. It is not an intact biological brain.

Incremental inference is also established. Stockfish’s NNUE documentation describes accumulator updates from sparse feature changes (Stockfish developers, 2026). InkStream and Ripple reuse computations in evolving graph neural networks (Wu et al., 2023; Naman and Simmhan, 2025). Our dependency-cone construction specializes this principle to branching chess successors and explicitly tests global-feature placement. It must be compared to those methods before any general efficiency claim. TypeSafe’s proprietary Jev is separate from the local research models; none of these experiments measures its internals or reconstructs its architecture.

3 EXPERIMENTAL ACCOUNTING

Inputs and reference targets. The shared spatial policy represents a board with nineteen channels and scores all native legal UCI candidates. The channels include piece ownership, castling, en-passant state and counters, in a side-relative rank orientation. No SAN mate markers or engine targets are supplied to the network. Decision labels use a 2,000-node Stockfish teacher. Higher-budget assessment uses Stockfish 19 with 20,000 requested nodes for each unrestricted or chosen-move-constrained call. For scores c_* and c_a in centipawns, the bounded loss is

$$L(s, a) = \tanh(c_*/600) - \tanh(c_a/600). \quad (1)$$

Finite-search scores can make this difference negative. It is neither exact game-theoretic regret nor a calibrated probability. Raw centipawn loss is retained in the underlying experiments where recorded, because the bounded transform can hide large changes at extreme scores.

Panels and uncertainty. Ordinary and shifted rollout panels vary the generator’s teacher/random-action mix and game length. Historical roots, legal successors and color mirrors are excluded according to each frozen plan. These are development distributions that have already influenced architecture choices. Candidate and biological experiments each evaluate 2,048 positions per panel, with fixed 128-position subsets for stronger-engine grading. Within an experiment, all arms use the same positions. Positions within games are dependent. Three fitted seeds are shown without treating them as a large population sample.

Training and total cost. The candidate comparison uses twelve fits: four architectures and three paired seeds, each with 32,768 training positions, six epochs and 1,536 Adam updates. The biological study trains eighteen adapters on three frozen direct-policy backbones with the same data count and updates. Backbone pretraining remains a cost of the biological study; this is not evidence of low-data efficiency. All final adapters finish before any neural evaluation. No best epoch, seed, replacement fit or retrospective budget extension is selected.

4 EXPLICIT CONSEQUENCES DO NOT ESTABLISH BETTER DECISIONS

The four candidate arms share a spatial root encoder, recurrent core and legal candidate readout. Direct scoring uses root features only. Action-only refinement injects source/destination/promotion

Table 1: Candidate experiment. All three seeds averaged within each arm. Agreement is percent; bounded engine loss is lower-is-better. These panels differ from Table 2.

Arm	Agree., ordinary	Agree., shifted	Loss, ordinary	Loss, shifted
Direct	31.75	26.81	0.09878	0.08163
Action only	31.54	26.45	0.09115	0.09525
Exact input delta	33.15	27.10	0.09283	0.08768
Full afterstate	32.19	27.10	0.10821	0.09537

features. Delta refinement also projects the exact nineteen-channel input difference between root and native successor. Full-afterstate refinement separately encodes the successor. The latter three share two branch recurrence steps after the four-step root computation. The native transitions are exact rules, not learned dynamics.

Delta improves mean bounded loss relative to direct by 6.02% on ordinary positions, but worsens it by 7.42% on shifted positions. Neither comparison meets the required 20% reduction on both panels; the action-only comparisons also fail. In 96 color-paired games per opponent, delta scores lower-bound point shares of 50.00% against direct and 50.52% against action-only, below the required 60%. Two of the 288 games remain unfinished and stay in the denominators. There are no failed games. Game scores are not converted into Elo.

Permuting which successor belongs to which candidate reduces teacher agreement on both panels in every seed. This intervention demonstrates dependence on the action-successor association. It does not show useful planning; that stronger claim fails the quality and gameplay tests. Delta’s complete decision latency is about 5.5 times direct scoring in the original local measurement, including native transition and input preparation. This cost motivates the locality hypothesis, without assuming the existing delta method has a quality advantage worth preserving.

External distribution check. All twelve frozen candidate models were evaluated without additional training on a selected 4,096-position public ChessBench panel. Mean reference-move agreement ranges from 26.03% to 26.79%. This external-source measurement is development evidence; source-game identifiers are absent, and it is not a sealed test or an independent-game uncertainty estimate. The entire 62,561-row acquired file is reserved in the later biological study’s exclusions.

5 CONTROLLED BIOLOGICAL TOPOLOGY COMPARISON

The audited packed graph has 138,639 nodes. A label-independent coarse-group rule selects all 1,409 descending nodes, inducing 44,090 signed connections and discarding 430,912 boundary connections. The selected graph contains six isolated nodes. Original packed node indices are the available identities; they are not asserted to be FlyWire root IDs or functional cell-type labels.

The adapter maps frozen board features into graph state with a fixed seeded mapping, performs four leaky scalar recurrence steps, and applies a learned residual correction through a zero-initialized output projection. State resets on every decision. Biological wiring is compared with three independently rewired graphs that preserve signed in/out degrees, dense recurrence, node-local recurrence, and the unchanged backbone. Each rewire completes 440,900 accepted swaps; structural audits also record final edge overlap. Biological and rewired arms match parameter count and operations. Dense and node-local references do not match those counts and are reported as additional controls.

The original criterion fails: 5 of 30 constituent checks pass. Biological mean loss reductions against individual rewires range from -12.09% to +0.29% across the two panels. These are descriptive paired comparisons, not a population-level significance test.

The frozen continuation rule requires at least 10% lower mean bounded loss against each rewire on both panels, a strict improvement in every paired seed, and no paired degradation from the unchanged backbone. Missing or nonfinite results cannot pass. This stage contains no arena or Elo estimate. A pass would justify a separately frozen gameplay and mechanism replication, not establish novelty. Every topology uses dense matrix operations in the implementation; sparse wiring must not be reported as sparse-kernel speed.

Table 2: Biological-topology experiment. Every arm and all three seeds retained. Loss is a signed bounded finite-engine score difference; agreement is percent.

Arm	Loss, ordinary	Loss, shifted	Agree., ordinary	Agree., shifted
Frozen direct	0.09739	0.11009	33.02	24.37
Biological	0.10198	0.10396	33.46	24.63
Rewire 151	0.09539	0.09741	33.35	24.32
Rewire 163	0.09099	0.10426	33.63	24.64
Rewire 179	0.09664	0.10266	33.41	24.45
Dense	0.09361	0.10694	33.45	24.56
Node-local	0.09250	0.10490	33.45	24.35

6 FACTORIZED COUNTERFACTUAL LOCALITY

6.1 ARCHITECTURE

The new prototype separates local and global state. Local features comprise twelve piece/color channels, rank/file coordinates and an en-passant marker. Two row-normalized directed adjacency matrices encode attacks by root-side and opponent pieces. Pinned pieces retain geometric attacks, and friendly occupied endpoints are included. These relations are derived from chess rules, not a biological connectome. Global metadata comprises side to move relative to the root player, four castling rights, two counters and en-passant availability.

Let X denote local features, g global metadata and A_r the adjacency for relation r . With biases implicit, the model is

$$z = XE, \quad h^0 = \tanh(z), \quad (2)$$

$$h^{t+1} = \frac{1}{2}h^t + \frac{1}{2}\tanh\left(h^t W_0 + \sum_r A_r h^t W_r + z\right), \quad (3)$$

$$q(s) = \text{MLP}[\text{mean}(h^T), g]. \quad (4)$$

Node features are row vectors. A broadcast control adds the same projected global vector gG to every row of z , including initialization. Global changes therefore invalidate all nodes in that control. In the factorized model they affect the readout immediately but need not invalidate local recurrence. The functions differ across architectures; only full and incremental evaluation of the *same* architecture must agree. Late fusion may also reduce modeling capacity, which remains to be measured after matched training.

6.2 EXACT DEPENDENCY-CONE UPDATES

Cache root states $h^0, \dots, h^T$ once. For a native successor in the same root-player perspective, let S_0 contain changed local input nodes and D contain destinations with changed normalized adjacency rows. Recursively define

$$S_{t+1} = S_t \cup S_0 \cup D \cup \{v : \exists u \in S_t, r, A'_{r,vu} \neq 0\}. \quad (5)$$

Recompute layer $t + 1$ only at destinations in S_{t+1} , using the updated predecessor layer, and copy other nodes from root layer $t + 1$.

Equivalence. Assume fixed parameters, deterministic node-local nonlinearities, and no hidden global normalization or pooling inside recurrence. At initialization, unchanged input nodes have identical states. Suppose all child states outside S_t equal their root states. For v outside S_{t+1} , the local input, self state and normalized adjacency rows are unchanged, and no child edge reaches v from a changed predecessor. Thus its entire update equals the root update. Induction proves equality at every layer in real arithmetic. Deleted edges and changed normalizers are included in D , preventing incorrect stale-cache reuse. This is an application of incremental-computation reasoning, not a claim of a new general theorem. Floating-point implementations require tolerance checks.

Table 3: Untrained locality screen. Logical recurrent-node updates include root caching, as a percentage of full recomputation. Errors are maxima over all candidates and three seeds, not measures of chess quality.

Architecture	Depth	Node updates (%)	Max. hidden error
Factorized	2	28.01	1.19e-07
Factorized	4	34.49	1.42e-07
Factorized	8	39.37	1.79e-07
Broadcast	2	103.15	0.00e+00
Broadcast	4	103.15	0.00e+00
Broadcast	8	103.15	0.00e+00

6.3 ENGINEERING SCREEN

Before inspecting outcomes, the screen fixed 64 old training roots in a label-independent hash order, all legal successors, widths of 32, depths 2/4/8, three initialization seeds, and factorized/broadcast arms. No optimizer, engine or external model is called. The screen checks every successor’s hidden state and scalar score against full recomputation with absolute tolerance 2×10^{-6} . A stale-root diagnostic distinguishes correct updating from doing no work.

The logical work comparison charges root-cache construction as well as all child node updates. Complete timing uses sixteen roots, three paired repeats, depth four and seed 211, alternating method order. It includes FEN parsing, legal moves, native board copies, graph construction, root caching where needed, propagation and readout. The current dense masks and tensor clones remain in the timed path. Fewer logical updates are not assumed to imply lower latency. The initial full baseline processes one candidate at a time. Its original gate remains unchanged after the following stronger baseline comparison.

The median paired incremental/full latency ratio is 1.068 for the factorized network and 1.757 for the broadcast control. The combined engineering criterion fails. No training or game-quality evaluation is performed. Logical savings and numerical equality do not establish a useful learned policy.

Batched baseline follow-up. A separately frozen implementation stacks all sibling graph inputs and uses batched dense multiplication to compute the same network. Nine unit tests check hidden states, outputs, gradients, sibling order, and special moves. The follow-up uses the same 64 already exposed roots, three depths, three seeds and both arms. Complete timing charges graph encoding and validation, stacking, recurrence, readout and conversion to Python scores, alongside the original full and incremental paths. Six method permutations rotate over sixteen roots and three repeats per arm at depth four and seed 211. No test positions or training labels are introduced. All 36,594 batched/single numerical comparisons pass the same tolerance; the largest hidden discrepancy is 3.50×10^{-7} and the largest score discrepancy is 1.19×10^{-7} . Median paired batched/full latency ratios are 0.361 (factorized) and 0.355 (broadcast). The original incremental implementation takes 2.906 and 4.812 times the batched baseline, respectively. Its separately fixed requirement of at least 10% lower latency fails. These shared-host development measurements establish no hardware-general speed ratio.

7 TWO-ENDPOINT RELATION TRANSPORT

The earlier candidate comparison motivates a different question: can a move head use long-range root-board relations without recomputing a spatial network for every successor? We implement two-endpoint transport as a controlled representation hypothesis. It uses no afterstate, teacher shortlist or search.

Mechanism. Freeze the three direct-policy backbones from the candidate experiment. For each legal move, initialize a width-32 query from its existing source, destination, pooled-board and promotion/displacement features. Initialize two distributions p^0 and r^0 as point masses at the move’s endpoints. Four row-stochastic matrices P_k represent mover attacks, their reverse, opponent attacks and their reverse. Pinned pieces retain geometric attacks; empty rows become self-loops. At each of

Table 4: Transport-head experiment on the existing candidate panels. All three paired backbones retained. Agreement is percent; target negative log likelihood is lower-is-better. These are development teacher-imitation measurements, not engine grading or game scores.

Head	Agree., ordinary	Agree., shifted	NLL, ordinary	NLL, shifted
Frozen direct	31.75	26.81	2.435	2.635
Full transport	33.02	27.99	2.358	2.575
No overlap	32.54	28.09	2.380	2.580
Uniform routing	32.49	27.34	2.390	2.602
Static endpoints	32.31	26.87	2.419	2.621
Rewired relations	32.19	27.70	2.391	2.588

three internal steps, the query produces separate four-way softmax gates α_k^t and β_k^t :

$$p^{t+1} = \sum_{k=1}^4 \alpha_k^t p^t P_k, \quad r^{t+1} = \sum_{k=1}^4 \beta_k^t r^t P_k. \quad (6)$$

Pool frozen node features under both distributions and their normalized pointwise product, retaining the product’s mass as another scalar. A shared GRU updates the query from these pools. A zero-initialized scalar output adds a correction to the backbone logit. All arms therefore start with identical predictions. The head stores 16,744 parameters; dense batched operations process the legal menu. The overlap represents root-graph accessibility, not consequences created by the move. Unattacked destinations may be absorbing.

Attribution controls and budget. Five heads compare full transport, zeroed overlap pool/mass, fixed uniform relation gates, stationary endpoint distributions, and full transport on degree-preserving rewired root graphs. The unchanged backbone is an additional reference. Every color-specific rewire preserves each node’s incoming and outgoing degrees, forbids duplicate/self edges, and requests five accepted swaps per edge under a fixed attempt ceiling. The prepared-input audit verifies all 32,768 training roots and 65,536 rewired relations. Every training relation reaches its swap target and changes; about 22.7% of original edges remain. This sampler is not uniform over all degree-matched graphs, and only one fixed FEN-derived rewire is used per root.

Each of fifteen heads trains for six epochs and 1,536 updates on the same 32,768 old training roots, with paired seeds, minibatches, Adam settings and legal-move cross-entropy. Backbones remain frozen; their pretraining cost is retained. All fits finish before evaluation. Controls store equal parameters, but inactive gates/input coordinates and different operations preclude claims of equal active capacity or FLOPs. Both evaluation panels are already exposed development data. The fixed continuation rule requires mean agreement gains of one point over uniform/static/rewired, half a point over no-overlap, and no loss against the backbone on both panels, with no paired-seed deficit worse than half a point. This teacher-imitation screen cannot establish engine-quality or gameplay improvements.

The fixed continuation rule fails: 3 of 10 panel/comparator checks pass. Transport’s mean agreement changes relative to the unchanged backbone are +1.27 and +1.19 percentage points on ordinary and shifted panels. Relative to the overlap ablation, they are +0.49 and -0.10 points. Local median complete-decision latencies are 0.435 ms for the backbone and 0.842 ms for full transport. The saved-record audit replays all 86,016 evaluation predictions, including test-time topology corruptions, and independently rebuilds the metric arithmetic and original criterion. No training replay, game evaluation or untouched confirmation is performed; a separate exploratory engine assessment follows.

Exploratory engine characterization. After the failed primary criterion was observed, we froze a separate assessment of all eighteen policies on the original candidate study’s 128-position subsets per panel. There is no model or position selection from the new agreement outcomes and no replacement success threshold. Each root receives a fresh unrestricted call and one call restricted to each unique policy choice, clearing the engine hash each time and requesting 20,000 nodes. All three paired seeds are retained, including finite-search negative differences. The separate assessment uses 817 fresh engine calls and 16,340,000 requested nodes, with no new neural inference. Full transport

Table 5: Post-hoc engine characterization, frozen after the primary transport gate failed. All eighteen policies use the original 128-position subsets per panel and fresh 20,000-node calls. Both signed bounded loss and raw centipawn loss are lower-is-better; finite-search differences can be negative.

Head	Bounded, ordinary	Bounded, shifted	CP, ordinary	CP, shifted
Frozen direct	0.09878	0.08163	642.6	428.6
Full transport	0.06535	0.07278	619.7	451.8
No overlap	0.07326	0.08137	587.2	433.1
Uniform routing	0.08893	0.07777	557.4	500.7
Static endpoints	0.09678	0.09533	653.6	454.1
Rewired relations	0.08286	0.07056	600.7	399.4

Table 6: Additional fixed-backbone head comparison, using all three seeds and the same old development panels. Agreement is percent. The first two rows are authenticated earlier results; three new heads train with the same examples and update budget. Head counts exclude the shared backbone.

Head	Head parameters	Agree., ordinary	Agree., shifted
Frozen direct	0	31.75	26.81
Full transport	16,744	33.02	27.99
Generic MLP	16,240	32.11	27.31
NCN adaptation	16,380	32.44	27.28
Typed overlap	16,144	32.45	27.18

reduces mean bounded loss from the backbone by 33.84% on ordinary positions and 10.84% on shifted positions. Raw centipawn loss, however, increases by 5.40% on the shifted panel. The rewired control has lower shifted bounded loss than full transport. The bounded transform saturates at extreme scores, while raw centipawn loss depends on the arbitrary mate-score encoding; these conflicting endpoints prevent a uniform quality-improvement claim. This post-hoc characterization does not alter the failed primary criterion and is not independent confirmation.

Simpler-head follow-up. A separately frozen nine-fit comparison adds a generic move MLP, an NCN-inspired endpoint-product/common-neighbor head, and a typed overlap head. All use the same frozen backbones, action features, old training data and 1,536 updates. The generic head has two 80-unit hidden layers; the untyped overlap head uses 65. The typed head pools all sixteen ordered pairs of attack/reverse-attack relations, projects each pooled vector from 32 to 8 dimensions with shared weights, and uses two 48-unit hidden layers. Head counts are within 4% of transport. All residual outputs start at zero. No test outcomes determine these widths. Untyped pooling uses an undirected attack union; typed pooling preserves direction and color. These are chess adaptations of structural pooling, not reproductions of the original NCN encoder and benchmark.

All nine fits finish before evaluation. Recomputing every training-root cache for all three backbones matches the original cache exactly. Original backbone and transport predictions are copied with authenticated provenance; they are not counted as fresh inference. The additional rule requires transport to gain at least one agreement point over each new head on both panels, with no paired seed deficit worse than half a point. The additional transport-advantage rule fails: 0 of six comparisons pass. Transport minus generic-head agreement is +0.91 and +0.68 percentage points on the two panels. The audit replays all 36,864 new baseline outputs and 24,576 copied reference outputs. Every paired-seed agreement difference favors transport, but the fixed minimum effect is not reached. This comparison does not change the earlier failed criterion and is not fresh confirmation. A separately frozen post-hoc engine extension retains all fifteen policies; Appendix B reports the mixed endpoint ranking.

Joint representation training. A separately frozen follow-up allows gradients through the root encoder, recurrent core, action embeddings and direct policy head. The five transport variants and a direct policy start from the same original paired backbones; transport heads are fresh and initially output zero. Unused value, auxiliary and action-plane projection modules remain fixed. Every arm receives six more epochs of legal-move cross-entropy at the same fixed learning rate and minibatch order. Eighteen final fits total 27,648 updates, with all training completed before evaluation. The

Table 7: Joint-training follow-up. All three seeds retained. Agreement is percent; NLL is lower-is-better. The direct comparator receives the same additional cross-entropy training as all transport variants.

Policy	Agree., ordinary	Agree., shifted	NLL, ordinary	NLL, shifted
Frozen direct	31.75	26.81	2.435	2.635
Joint direct	32.01	27.23	2.414	2.625
Joint transport	33.53	27.25	2.354	2.571
Joint no overlap	33.11	27.49	2.362	2.583
Joint uniform	32.99	26.95	2.379	2.610
Joint static	32.11	26.90	2.406	2.628
Joint rewired	33.37	27.85	2.370	2.578

direct comparator controls for additional backbone training; parameters and FLOPs are not equal across arms.

The fixed criterion requires at least one agreement point over jointly trained direct/uniform/static/rewired, half a point over no-overlap, and no loss against the unchanged backbone on both old panels. Every paired-seed deficit must be at most half a point. This tests a fixed optimization budget, not individually tuned learning rates or an untouched confirmation distribution. The joint-training criterion fails: 4 of twelve checks pass. Joint transport minus jointly trained direct agreement is +1.51 and +0.02 percentage points on ordinary and shifted panels. Relative to the preceding frozen-representation transport-minus-base margin, this changes the margin by +0.24 and -1.17 points. The audit checks all 27,648 training records and replays all 98,304 evaluation predictions, including graph corruption. Encoder and policy weights change while unused auxiliary/value/projection parameters remain fixed. This matched-update optimization comparison does not establish a new transport mechanism, fresh confirmation or gameplay improvement.

Prior-work boundary. Learned relation-operator mixtures are established in Neural LP (Yang et al., 2017); NBFNet constructs source/query-conditioned pair representations through learned path aggregation (Zhu et al., 2021). Neural Common Neighbor uses structural overlap to guide learned feature pooling (Wang et al., 2024). The present head combines related ingredients for chess decisions. A separately frozen NBFNet-style adaptation exceeds transport by 1.51 and 0.91 agreement points on ordinary and shifted panels (Table 9). Transport passes 0 of its two additional advantage checks; all 36,864 baseline/reference predictions replay. These comparisons do not establish a new algorithm or a uniform quality advantage.

8 WHAT REMAINS BEFORE THE MAIN CLAIM

The experiments delimit failure modes and provide executable hypotheses, not a general quality-cost frontier. A decisive next study must identify a mechanism that survives the closest published-method comparators and task-specific interventions. The completed graph comparisons favor the established WLDN adaptation over scalar contrast (Appendix C); child-assignment sensitivity is insufficient. Controls need explicit information, training and inference budgets. A matched pin-factor follow-up passes only 2/16 checks: joint pin processing improves over the frozen backbone but has lower mean agreement than every trained control on both exposed panels (Appendix G). This does not establish a novel mechanism advantage. Efficiency claims also require suitable NNUE and incremental-GNN baselines. Architecture choices must be frozen before a disjoint confirmation panel is exposed.

A further hypothesis is a learned board-to-connectome mapping compared to identically learned mappings on several degree-preserving rewires. It is not implemented as a validated improvement here. Combining the tested transport, locality and biological modules would require factorial ablations showing that each contributes beyond the others.

Longer-term claims need strong engine-quality and paired-game evidence with source-game-aware uncertainty, plus a second relational task if the claim is intended to generalize beyond chess. A successful engineering screen alone cannot meet those requirements. Human scientific review, complete citation verification, and an anonymity audit remain prerequisites to submission.

REPRODUCIBILITY STATEMENT

Frozen protocols bind code, input files, checkpoints, seeds, exclusions and environments. Completed runs retain all predictions, costs, unsuccessful arms and unfinished games. Tables in this draft are generated from hash-verified saved summaries. The biological report separately audits raw outputs without new inference. Local artifact identities accompany the development source; an anonymous, distributable submission package has not yet been prepared. External biological assets and derivative weights remain local pending their applicable source terms. Original graph-control code uses synthetic test inputs.

AI USE STATEMENT

An OpenAI coding assistant contributed literature discovery, architectural hypotheses, implementation, tests, experiment orchestration, analysis and drafting. This draft and its claims have not yet completed human scientific review. Authors must verify the results, attribution, originality and final text before any submission. No model is listed as an author. This statement must be reviewed against the final contributions and submission-form disclosures.

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A EARLIER DEVELOPMENT EXPERIMENTS

The initial synthetic circuit had mean teacher agreement of 14.36%, versus 14.19% for a signed degree-preserving rewire, below its required three-point advantage. A subsequent spatial study compared untied convolution, tied recurrence, current-board reconstruction and future-piece prediction. Prediction learned changed-square targets but failed the prescribed decision-quality gain over the controls. Mixed-depth and anchored recurrence were examined separately; they did not establish the intended robust extra-computation benefit.

A six-fit width comparison increased active parameters from 33,185 to 439,073. Its mean bounded engine loss improved by 22.99% on ordinary positions and 9.26% on shifted positions, missing the required 20% improvement on both. Against the small model it scored 18 wins, 64 draws, 12 losses and two unfinished games, below the required lower-bound 60% point share. These experiments used different panels and budgets from the main tables and must not be pooled into a single benchmark or used as independent replications.

B STRONGER-ENGINE CHARACTERIZATION OF SIMPLER HEADS

After observing the additional teacher-agreement comparison, we froze another post-hoc engine assessment. It includes every paired backbone and full-transport policy plus all nine newly trained simpler heads. The original 128-position subsets remain fixed; no policy or position is selected from favorable new outcomes. Every unrestricted and unique chosen-move call starts with a cleared hash and requests 20,000 nodes under the same single-threaded Stockfish setup.

Table 8: Post-hoc engine assessment of all fifteen simpler-head policies on the same fixed 128 positions per panel. All calls are fresh and request 20,000 nodes. Both loss measures are lower-is-better; raw centipawn loss includes an artificial mate-score encoding.

Head	Bounded, ordinary	Bounded, shifted	CP, ordinary	CP, shifted
Frozen direct	0.09878	0.08163	642.6	428.6
Full transport	0.06535	0.07278	619.7	451.8
Generic MLP	0.09502	0.08370	650.9	437.2
NCN adaptation	0.08712	0.08185	601.6	460.4
Typed overlap	0.08773	0.08364	600.7	423.3

The new assessment uses 791 fresh calls and 15,820,000 requested nodes. Full transport has lower mean bounded loss than each simpler head on both panels. Raw centipawn loss favors typed overlap on both panels and untyped overlap on the ordinary panel. Thus the mean bounded-loss ranking survives these simpler controls, but there is no uniform ranking across endpoints. The audit checks all 3,840 policy-position records, and fresh backbone/transport means exactly match the preceding engine assessment. No new neural inference or gameplay is performed; this post-hoc extension cannot replace either failed primary criterion.

Published path-baseline adaptation. A separate three-fit study uses a 16,416-parameter NBFNet-style head: a 120-to-16 action query, three width-16 DistMult/PNA layers with source boundary reinjection, residual shortcuts and layer normalization, followed by a target lookup and width-32 readout. The zero-initialized scalar correction preserves the same original backbones. It uses the same four typed root attack channels, 32,768 roots, six epochs and 1,536 updates per seed; all fits precede evaluation. Selected pinned public unfused methods pass value/gradient checks through a scatter compatibility layer. This is a documented chess adaptation, not an official NBFNet benchmark reproduction. Unlike transport, it accesses frozen node features through the endpoint/global query rather than pooling features of every visited node. Similar stored parameters do not equalize compute.

Table 9: Separately frozen path-baseline comparison. Agreement is percent; NLL is lower-is-better. All three paired seeds retained; original references copied with authenticated provenance.

Head	Agree., ordinary	Agree., shifted	NLL, ordinary	NLL, shifted
Frozen direct (copied)	31.75	26.81	2.435	2.635
Transport (copied)	33.02	27.99	2.358	2.575
NBFNet-style path head	34.54	28.91	2.298	2.514

The original transport needed at least one agreement point over this comparator on both panels, with no paired-seed deficit worse than half a point. Both checks fail. The audit verifies all 4,608 update-index records and replays 12,288 new baseline predictions plus 24,576 copied references. The original failed transport criteria remain unchanged; there is no new engine or gameplay result. Paired complete-decision medians are 0.586, 1.106 and 5.243 ms for the backbone, transport and path head, respectively. The path head takes 4.74 times transport’s median in this rotating shared-host measurement; the quality comparison is not a matched-FLOP or matched-latency result.

C CANDIDATE-SPECIFIC ATTACK OPERATORS

Root-graph transport omits changes to attack lines caused by a proposed move. An untrained follow-up computes four child operators in the fixed root-player frame, using native transitions for all legal candidates. Write $D_{a,r} = P_{\text{child}(a),r} - P_r$, including all changed normalization coefficients and empty-row self-loops. Sparse indexed accumulation implements $pP_r + pD_{a,r}$. This is the linear identity for applying the child operator, not a new theorem or learned transition model. Both paths first reconstruct native child graphs; the prototype is not an incremental attack generator. Nine tests cover color frames, castling, en passant, promotions, dense operator reconstruction and three-step output/gradient equivalence.

The frozen screen uses point masses at the move endpoints and seeded random gates, node features and readout weights. It has no trained query controller or chess-quality evaluation. Complete untrained pipeline timing includes board parsing, legal enumeration, graph construction, normalization/delta packing, input preparation, three propagation steps, readout and UCI selection. It uses sixteen roots, three paired timing repeats and a shared host. The untrained screen covers 7,623 candidate/step comparisons over 32 old training roots and three seeds. Maximum distribution error is 3.58×10^{-7} , but maximum readout error is 3.81×10^{-6} , above the fixed absolute tolerance of 2×10^{-6} . Median sparse/dense operator-storage ratio is 0.0675, with lower storage on every root; this is not peak memory. Median paired complete-pipeline latency ratio is 1.410, failing the required ratio of at most 0.90. The original numerical and combined engineering criteria fail. All output/storage records replay; timings are authenticated rather than repeated. Any later repair must

preserve this failed screen. A quality study would first need to establish a benefit from the full child-operator representation against root-only and candidate-assignment controls.

Shared-weight graph contrast. A separate dense prototype scores the change in a candidate-conditioned graph representation. With frozen root features H_s , action features $q_s(a)$ and the same learned head f_θ in both branches, its correction is

$$c(s, a) = f_\theta(P_{s,a}; H_s, q_s(a)) - f_\theta(P_s; H_s, q_s(a)). \quad (7)$$

The child graph comes from the exact native move in the root-player frame; node features remain those of the root. Shared controllers may diverge after receiving different pooled contexts, so this is not the fixed-gate linear operator identity above. A zero-change mask enforces exact cancellation in both forward and backward execution when the operators agree. The root term is action-dependent, so subtracting it can affect the legal-move ranking.

The four implemented arms are this difference, child only, root only, and a label-independent permutation of child-to-candidate assignments. Each stores 16,744 parameters and starts with zero residual output. Eighteen tests include special moves, direction, parameter identity, separate-branch equivalence and menu alignment. The first screen stopped on a tuple/list validation mismatch; its failed receipt is retained. A separately frozen repair changes that validation, retaining all positions, seeds and thresholds. The numerical screen passes over 96 root/seed checks and 2,628 candidate checks. Maximum root, child and score-difference discrepancies are each at most 4.77×10^{-7} against a fixed 2×10^{-6} tolerance. Identical operators produce exactly zero correction and parameter gradient. Twelve artificial-target optimizer updates check gradient flow and cost; their complete checkpoint replay is exact. Those twelve audit updates are additional computation. No development teacher agreement, engine score or game result is measured for this head. A separately frozen implementation shares root propagation and retains compact legal-child operators. Its 96 root/seed and 2,628 candidate checks pass with maximum discrepancy 4.77×10^{-7} . Exact zero-change behavior and twelve additional artificial-update checkpoint replays also pass. A lossless bit cache covers 36,864 existing roots and 1,087,523 legal children; every stored graph is independently reconstructed. These engineering checks establish neither trained quality nor an inference-speed advantage.

Completed scalar-contrast quality study. The completed twelve-fit study retains 18,432 updates and fails its original criterion: 4 of eight checks pass (Table 10). Contrast minus child-only agreement is +0.16 points ordinary and -0.20 shifted. It improves over the base and trained shuffled-child control but does not establish the required advantage over root-only or child-only scoring. All 73,728 saved predictions replay with zero NLL discrepancy; all ten descriptive bootstrap intervals are recomputed. Root-only checkpoints exactly reproduce original transport for all three seeds.

Table 10: Completed scalar-contrast study on ordinary (O) and shifted (S) panels. Agreement is percent; NLL is lower-is-better. All three seeds retained. Corrupted contrast reuses fitted contrast weights; it is a diagnostic, not a new fit.

Head	Agree. O	Agree. S	NLL O	NLL S
Frozen base	31.75	26.81	2.435	2.635
Root only	33.02	27.99	2.358	2.575
Child only	33.25	28.14	2.347	2.573
Scalar contrast	33.41	27.95	2.364	2.571
Trained shuffled contrast	32.06	26.33	2.432	2.637
Test-time shuffled contrast	31.33	26.68	2.458	2.647

An input-only audit and independent replay cover all 1,087,523 cached legal candidates. Every candidate changes the graph and vacates a nonempty own-attack source row; no candidates within a root have identical child graphs. Stationary pieces change attack lines for 62.04%, 60.62% and 59.91% of training, ordinary and shifted candidates, respectively. Thus the exact-zero-change invariant is synthetic on these panels, and a child-assignment shuffle can disrupt simple move/source consistency as well as tactical relations. No fitted-model reliance on either signal is established. A separate input-only screen over 128 old training roots finds nonzero inverse-channel mixed differences from separate normalization on 1,157 of 3,920 legal candidates; a fixed untrained linear

projection detects 1,146. Independent native-graph and NumPy reconstruction replays every value exactly. Thus an interaction response can arise without learned nonlinear scoring. A common union denominator with diagonal slack cancels the single-step mixed difference to numerical precision, but changes the information path and computational cost and is not a demonstrated policy improvement.

Paired chess representations already use shared weights (David et al., 2017). More directly, WLDN ranks candidate reaction outcomes using differences of graph representations and adds its score to an earlier proposal score (Coley et al., 2019). Shared graph encoders with subtractive atom representations also predict activation energies (Grambow et al., 2020), while condensed reaction graphs can encode changes directly in node and edge features (Heid and Green, 2022). Our action-conditioned transport and scalar subtraction differ from their nodewise difference encoders; this distinction alone is not a contribution. The ingredient controls do not replace the direct method comparisons below. Perturbed-graph explanation is another established use (Lucic et al., 2022).

Direct prior-method adapters. We implement two additional baselines to separate scalar-score differences, nodewise differences and changes encoded directly on the input graph. The WLDN-style adapter (Coley et al., 2019) applies a shared width-32 encoder with three tied WL updates to root and child graphs, subtracts the node states, and applies one separate WL update on the child graph before summing over squares. It concatenates the result with the original 120 action features and uses a 59-unit readout, for 16,638 head parameters. Edge features and biases in the difference encoder mean identical branches need not have zero learned scores. We retain that property rather than imposing scalar-contrast cancellation.

The condensed-graph adapter (Heid and Green, 2022) uses the union of root and child attack edges. Each message direction carries four root direction/color flags and four signed child-minus-root flags. A directed MPNN excludes immediate reverse-edge messages and adds the original edge projection at each of two tied recurrent updates. Width is 32, pooling averages all 64 square states, and an 80-unit readout receives the same action features, for 16,704 parameters. All node features remain those of the root in both adapters; the resulting zero node differences are omitted from the condensed input. These are explicit chess adaptations, not reproductions of molecular features or chemical tasks.

Selected pinned public functions pass output and gradient checks: WLDN through a Torch compatibility layer, and the Chemprop directed encoder through its unmodified Torch class with supplied graph tensors. The WLDN source and port retain GPL identification; the selected Chemprop source and adapter retain MIT identification. Neither check reproduces the full published training pipeline. Each baseline passes a separate native-input screen over 24 root/seed pairs and 627 candidates. Maximum separate-candidate versus batched score errors are 1.67×10^{-6} for WLDN and 2.62×10^{-6} for the condensed-graph adapter, below their predeclared 10^{-5} tolerance. Each preserves the initial backbone exactly and replays three artificial-target updates to an identical checkpoint. These tolerances do not change earlier failed screens. No teacher-agreement, engine or gameplay result is established by these preflights. A separately frozen three-seed WLDN study uses the same 32,768 roots, six epochs and minibatch order as scalar contrast. The WLDN study completes all three final fits and 4,608 updates; all 24,576 new/copied predictions replay with zero NLL discrepancy. Mean teacher agreement is 37.01% ordinary and 31.43% shifted, versus 31.75% and 26.81% for the frozen base. Mean WLDN NLL is 2.161 and 2.355. These are descriptive development results for an established-method adaptation; they do not establish a new architecture, engine strength or equal inference compute. A separate condensed-graph study completes the same three seeds and 4,608-update budget, with all 24,576 new/copied predictions replayed. Its protocol was frozen before inspecting final WLDN or graph-contrast quality outcomes, with NBFNet’s advantage over original transport already known.

Combined established-method comparison. The first separate comparison retains all eight original ingredient checks and adds two WLDN checks. A further frozen comparison adds two checks each for CGR and NBFNet. Every new check requires at least one point of mean contrast advantage and no paired-seed deficit beyond half a point. All fourteen checks are required; a later comparison cannot rescue an earlier failure. WLDN and NBFNet outcomes were known when the combined analysis was frozen, while scalar contrast and CGR were still training. This is adaptive development, not a preregistered research program.

Table 11: Identity-joined graph-baseline comparison on the same ordinary (O) and shifted (S) development panels. Agreement is percent; NLL is lower-is-better. Milliseconds are median complete native decisions in one balanced six-method process. Three paired seeds; similar parameter counts do not match inference compute.

Head	Agree. O	Agree. S	NLL O	NLL S	ms
Frozen base	31.75	26.81	2.435	2.635	0.452
Root transport	33.02	27.99	2.358	2.575	0.889
NBFNet-style	34.54	28.91	2.298	2.514	4.248
Scalar contrast	33.41	27.95	2.364	2.571	3.734
Nodewise WLDN	37.01	31.43	2.161	2.355	4.876
Condensed graph	34.08	28.74	2.333	2.542	3.962

The combined fourteen-check criterion also fails: 4 checks pass. WLDN exceeds scalar contrast by 3.60 and 3.48 agreement points on the two panels. The audit authenticates all four model replay audits, joins all 73,728 distinct method/seed/position records, retains all earlier failed checks, and recomputes six source-game intervals. These descriptive development comparisons favor an established nodewise-difference adaptation over the proposed scalar contrast. They provide no independent confirmation, engine-strength result or new algorithmic contribution.

D FIXED-WEIGHT WLDN PATHWAY DIAGNOSTIC

After inspecting the completed baseline comparisons, we froze a diagnostic of the strongest adapted head. It retains all three trained WLDN checkpoints and both complete 2,048-position development panels. No weights, labels or legal menus change. Five interventions separate the encoded child-minus-root node differences from the graph inputs to the subsequent difference layer.

Table 12: Fixed-weight WLDN pathway interventions on ordinary (O) and shifted (S) development panels. All three seeds retained. Agreement is percent; NLL is lower-is-better. These are interventions on trained heads, not separately trained architectures.

Intervention	Agree. O	Agree. S	NLL O	NLL S
Frozen base reference	31.75	26.81	2.435	2.635
Native WLDN	37.01	31.43	2.161	2.355
Zero node difference	31.56	26.25	2.436	2.647
Root graph at difference layer	36.30	30.00	2.204	2.411
Zero difference-layer edge flags	35.07	29.49	2.243	2.432
Zero pooled graph channels	31.25	26.68	2.432	2.623
Shuffle node differences	26.38	22.28	2.826	3.056

Zeroing node differences retains child adjacency, edge flags and learned biases. Replacing the difference-layer graph with the root graph retains the native encoded differences. Zeroing its four edge flags retains child adjacency, so this intervention does not remove all structural information. Zeroing the pooled graph channels retains the original 120 action features and trained readout. Finally, the frozen candidate permutation shuffles node differences within each legal menu while keeping the actual child’s graph and action features. The permutation can contain fixed points.

All 73,728 intervention/native predictions and 12,288 copied base references replay through the original head with temporary hooks; the native predictions reproduce the original study. Zeroing encoded node differences reduces mean agreement by 5.45 / 5.18 points on ordinary/shifted panels. Shuffling those differences within each legal menu reduces it by 10.63 / 9.15 points. Every native-minus-intervention agreement interval is positive under the declared source-game bootstrap; these ten descriptive intervals do not correct for adaptive development or multiple comparisons. The audit also verifies unchanged head states, removal of temporary hooks, native root observations, legal menus and labels. It authenticates the prior complete native child-graph audit rather than reconstructing that cache again. Bootstrap resampling uses 2,000 draws of source games after averaging paired correctness across the three seeds; the same draws serve all interventions within a panel.

The trained model depends on candidate-aligned node changes and on the subsequent graph context. These effects are not additive contributions: interventions can create out-of-distribution hidden states, and the difference layer combines node and edge information nonlinearly. In particular, the zero-difference condition is not a separately trained edge-only architecture. These results motivate retaining nodewise differences in the next controlled architecture comparison; they do not establish a new mechanism or predict the performance of retrained variants.

E UNION DIFFERENCE-LAYER HYPOTHESIS AND ENCODING LIMIT

An implemented extension retains the shared WLDN encoder (Jin et al., 2017) and nodewise difference but changes the graph used by the final difference layer. Four arms use child connectivity, root/child union connectivity, a union with explicit retained/added/removed relation flags, or the same union with corrupted edit labels. Each head stores 16,740 parameters with a width-32 encoder and a 58-unit action readout. The child/union controls leave eight of twelve edge input channels unused, so stored-parameter matching does not equalize active capacity. The four-arm engineering screen and audit cover 96 root/seed/arm checks and 2,508 candidate scores, including independent Python graph packing. The maximum batched-versus-individual score discrepancy is 1.91×10^{-6} under a frozen 10^{-5} tolerance. All twelve artificial updates and their final states replay exactly. These use first-legal-action targets and establish no chess-quality result.

The negative control rotates edit classes over ordered square pairs within each candidate and direction/color channel, preserving union flags and class counts. It can break reverse-direction consistency and leave homogeneous groups unchanged. A globally fixed added/removed swap would instead be a learnable input renaming. The separate quality study trains all twelve final models before evaluation, retaining the original WLDN and backbone as authenticated references. Its unchanged criterion requires an edits advantage over every trained comparator on both panels.

A separate input-only audit and independent NumPy replay cover all 36,864 cached roots and 1,087,523 candidates; the earlier native cache audit is authenticated, not rerun. The rotation changes every candidate and relabels 92.25% of added/removed direction/color instances. It also breaks reverse-direction class consistency in 99.98% of candidates. Thus this control is a broader corruption test: a performance difference cannot be attributed uniquely to correct edit placement. A reverse-consistent control would be required for that narrower claim.

Matched training result. All twelve fits and 18,432 updates complete; 2 of ten frozen checks pass, so the all-check criterion fails (Table 13). Edits minus original WLDN agreement is +0.21 points ordinary and +0.49 shifted, against a required one-point mean advantage on each panel. All 86,016 new/copied predictions replay, all minibatch-index receipts are checked, and all ten conditional source-game bootstrap intervals are recomputed. The maximum replay NLL error is zero. The descriptive 95% intervals for edits minus WLDN are [-0.64, +1.13] and [-0.12, +1.08] points; both include zero. These are adaptive teacher-imitation results on already exposed panels, not independent confirmation or measured game strength. Complete-native timing retains 1,728 decisions; all 288 distinct seed/root/method choices replay against the authenticated panel predictions. The edits/WLDN median paired ratio is 1.510. This descriptive same-process, shared-host measurement includes board parsing, legal menus, backbone evaluation, native child graphs and the head. It does not establish a hardware-general speed advantage or equal training/inference compute.

Reverse-consistent control. A separate implemented control rotates raw directed/color classes, then derives incoming flags by transposition. It preserves union support, per-candidate/channel class counts and reverse consistency. A bounded screen reconstructs 128 native training roots and independently checks all 3,920 candidates: every candidate changes, with zero reverse inconsistencies. There are 627 numerical score checks with maximum discrepancy 1.43×10^{-6} against a frozen 10^{-5} tolerance. The original head with independently packed flags replays all three artificial updates and the final checkpoint exactly. This control has no trained quality result; its artificial labels need not describe a legal chess successor. The original twelve-fit study retains its declared corrupted-label control.

Table 13: Union difference-layer development comparison. All three paired seeds retained. Agreement is percent; NLL is lower-is-better. Time is the median paired complete-native decision ratio to WLDN on sixteen fixed ordinary roots, with six rotating repeats. The test-time corruption reuses edits weights and has no separate timing.

Policy	Agree., ord.	Agree., shift	NLL, ord.	NLL, shift	Time / WLDN
Frozen direct	31.75	26.81	2.435	2.635	0.112
Original WLDN	37.01	31.43	2.161	2.355	1.000
Matched child	36.88	31.15	2.172	2.363	1.414
Union	36.70	31.72	2.163	2.365	1.437
Edit labels	37.22	31.92	2.153	2.356	1.510
Rotated labels	36.64	31.53	2.161	2.358	1.626
Edits, corrupted at test	36.36	31.36	2.167	2.366	–

Equivalent feature coordinates. For one binary relation, let $r, c, u = r \vee c$ denote root, child and union indicators, and let t, a, ℓ denote retained, added and removed indicators. Then

$$t = r + c - u, \quad a = u - r, \quad \ell = u - c; \quad u = t + a + \ell, \quad r = t + \ell, \quad c = t + a. \quad (8)$$

Thus these three-channel encodings are invertibly related for each of the four direction/color relations. An edge projection with weights W_t, W_a, W_ℓ has exactly the same preactivation in the presence basis when

$$W_u = -W_t + W_a + W_\ell, \quad W_r = W_t - W_a, \quad W_c = W_t - W_\ell. \quad (9)$$

All node columns, biases, connectivity and subsequent layers remain unchanged, so the complete scoring function is preserved in real arithmetic. Seven tests verify the Boolean truth table, inverse transforms, full-head scores and chain-rule gradients, including nonzero projections and hook cleanup. The float64 score/gradient tolerances are $10^{-12}/10^{-10}$.

This is an elementary reparameterization, not a new theorem or an expressivity advantage. It does not equate child-only with union connectivity. The transform is nonorthogonal, so isotropic random initialization, coordinatewise Adam and Euclidean gradient clipping need not produce equivalent training trajectories. Any basis-dependent training benefit would require a separate controlled comparison. Established condensed reaction graphs already encode both sides of a transformation (Heid and Green, 2022); naming edit classes is insufficient for an algorithmic novelty claim. BonDNet also forms learned atom/bond differences and represents a broken bond by a negative reactant feature before set-to-set pooling (Wen et al., 2021). Its readout differs from the final union-graph WL step here; that distinction alone does not establish novelty.

F SELECTED-PANEL OVERLAP SENSITIVITY

A post-hoc input audit connects source games sharing a selected root or a legal one-move successor, treating the color/rank mirror as equivalent and ignoring move counters while retaining literal en-passant state. The ordinary panel retains 39 components for 39 games, and the shifted panel retains 31 for 31; there are no cross-game links under this exposure definition. Largest groups contain 61 and 92 positions. Thus this particular overlap grouping does not change source-game bootstrap units. It does not certify whole-rollout separation, general statistical independence, seed-population uncertainty or protection against adaptive use of the development panels. No new positions, model predictions or engine calls are used for this audit.

G MATCHED ABSOLUTE-PIN WITNESS COMPARISON

The completed `chess-pin-quality-v3` study tests whether joint processing of rule-derived king, blocker and attacker witnesses improves candidate ranking beyond the established graph-difference path and simpler controls. It is *adaptive development on previously exposed panels*, not independent confirmation. All eight arms use the same authenticated canonical root features, legal menus, complete native child graphs and pin-factor inputs, with the information restrictions appropriate to each control. The designated treatment is the joint pin-factor head. Comparators are

WLDN, first-order separable and pairwise factor heads, root-only pin context, six owner/status pin counts, a graph MLP without witnesses, and the previously selected edit-labelled union head. The frozen backbone is evaluated separately for every seed.

Matched recipe, unequal computation. All 24 fits use seeds 97, 109 and 127, with 32,768 training roots, six epochs, batch size 128, and 1,536 Adam updates each. Every fit starts fresh and all training finishes before evaluation. The recipe fixes the learning rate at 10^{-3} , gradient clipping at one and legal-menu cross entropy; it does not select checkpoints by evaluation results. The stored head sizes range from 16,638 to 16,740 parameters. This proximity does not equalize active capacity, FLOPs or optimization. In particular, the pairwise restriction applies inside the anchored factor MLP, not to all policy interactions. Pooling a named motif does not itself establish a new architecture family.

Table 14: Matched pin-quality v3. Means over all three seeds; agreement is in percent and target NLL in nats. Each exposed development panel has 2,048 roots. All eight trained arms and all three frozen backbones are retained; the designated joint arm fails its quality criterion.

Method	Agreement (%)		Target NLL	
	Ordinary	Shift	Ordinary	Shift
Frozen backbone	31.755	26.807	2.434607	2.634750
WLDN	37.158	31.917	2.153503	2.348466
Joint pin	36.426	31.396	2.172513	2.368147
Separable	37.012	31.510	2.161526	2.358588
Pairwise	36.572	31.445	2.165616	2.348599
Root only	36.751	31.462	2.169414	2.366796
Counts	36.637	31.608	2.152418	2.352115
Graph MLP	36.572	31.657	2.166158	2.366070
Union edits	36.686	31.657	2.165127	2.360217

The original criterion fails. Only 2/16 checks pass, both against the frozen backbone. The joint arm has lower mean agreement than every trained comparator on both panels (Table 15). Its gains over the frozen backbone therefore do not establish a benefit of the proposed joint mechanism. The continuation criterion remains failed; WLDN cannot be substituted as the designated treatment after observing these results.

Table 15: All sixteen original joint-minus-comparator quality checks. Mean gain and worst paired-seed gain are percentage points. Every check requires a paired-seed floor of -0.5 points; mean gain must be at least zero against the backbone and $+1.0$ against trained controls.

Panel	Comparator	Mean gain	Worst seed	Check
Ordinary	Frozen backbone	+4.671	+4.102	Pass
Ordinary	WLDN	-0.732	-1.562	Fail
Ordinary	Separable	-0.586	-1.123	Fail
Ordinary	Pairwise	-0.146	-0.391	Fail
Ordinary	Root only	-0.326	-0.537	Fail
Ordinary	Counts	-0.212	-0.879	Fail
Ordinary	Graph MLP	-0.146	-0.586	Fail
Ordinary	Union edits	-0.260	-0.635	Fail
Shift	Frozen backbone	+4.590	+3.662	Pass
Shift	WLDN	-0.521	-2.100	Fail
Shift	Separable	-0.114	-0.391	Fail
Shift	Pairwise	-0.049	-0.928	Fail
Shift	Root only	-0.065	-0.439	Fail
Shift	Counts	-0.212	-0.732	Fail
Shift	Graph MLP	-0.260	-0.928	Fail
Shift	Union edits	-0.260	-1.270	Fail

Conditional uncertainty. The saved audit also recomputes all sixteen 95% source-game bootstrap intervals, using 2,000 draws and averaging the three paired-seed correctness differences within each

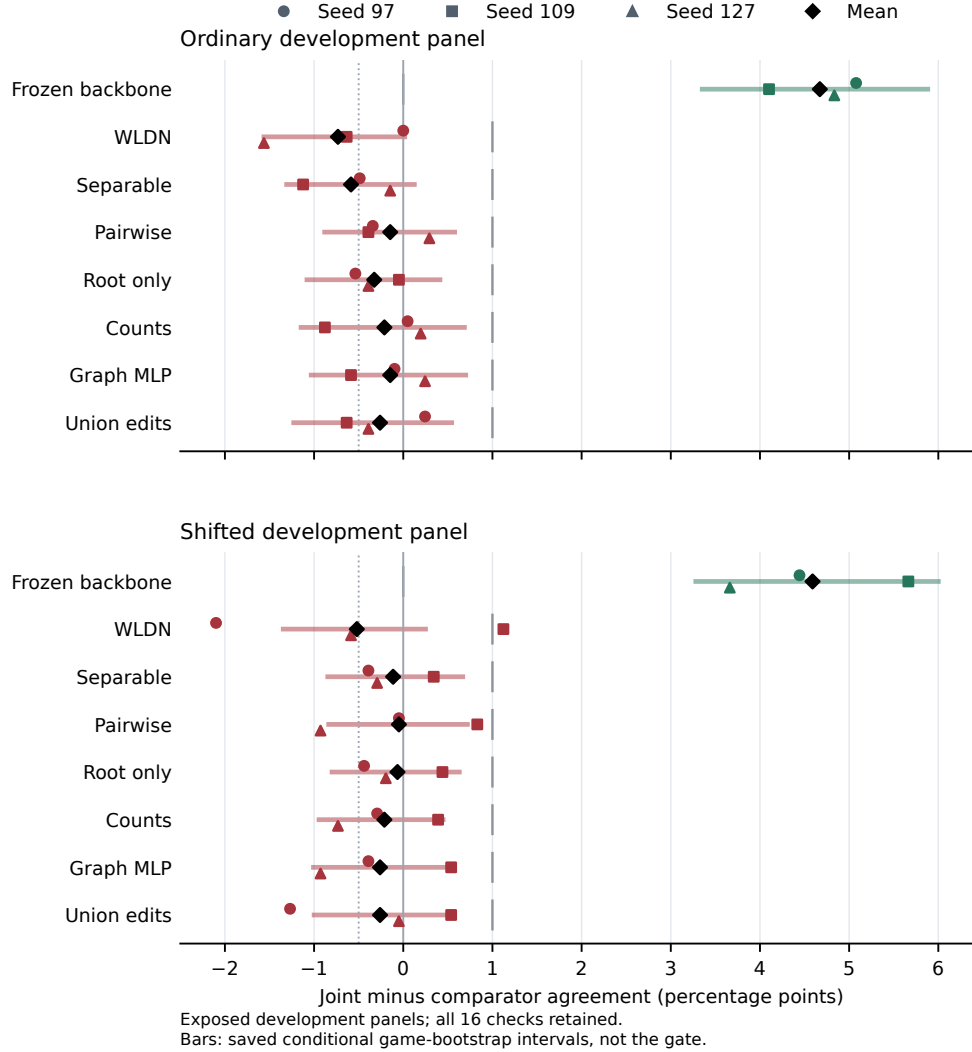


Figure 1: All paired seed differences for the original sixteen comparisons. The figure uses the authenticated completed audit report without new scoring or resampling. Symbols retain all three seeds; bars are conditional source-game bootstrap intervals. Green/red denotes the full check passing/failing. Short vertical ticks mark required means and the dotted line the paired-seed floor. These are previously exposed development data.

root before position-weighting sampled games. There are 39 source games in the ordinary panel and 31 in the shifted panel, each with 2,048 roots. These intervals describe the fixed trained models and exposed panels. They include neither uncertainty over new training seeds nor correction for adaptive development or multiple comparisons. They are not a substitute for the failed original continuation rule.

Audit scope and numerical agreement. The completed audit authenticates source, input and execution manifests, all 24 initial and final checkpoint identities, and 36,864 training-index receipts. It replays 110,592 prediction records and 3,226,149 legal-candidate scores, including 12,288 native root/backbone reconstructions. It recomputes all cached scores and target NLLs exactly, and reproduces native vectors and comparisons with the same production neural kernels. The native/cached numerical gate passes: the maximum score discrepancy is 3.8146973×10^{-6} , below 10^{-5} , with zero failed method-position cases and zero changed choices. This audit is not an independent neural

implementation or a repeat of training. The table and figure generators perform no new model or engine calls.

Elapsed costs and incomplete earlier attempts. Execution took 17,849.14 seconds and the separate audit 674.46 seconds. The 17,232.35 seconds summed over fit receipts are nested inside execution, not an additional cost. The earlier pin-quality-v2 attempt stopped before evaluation after 8,239 logged updates and five completed fits; its termination cause is unknown. Its 3,147.98 logged update seconds are a lower bound, and the overlapping 3,016.41 completed-fit seconds must not be added to that bound. Pin-quality-v1 stopped after 770 logged updates because a game-ID validator rejected integer IDs, with no completed fit or quality predictions. V3 restarts every fit; it reuses no earlier partial checkpoint. These inputs do not quantify all historical project costs. Shared-host elapsed times are not isolated inference latency or throughput measurements.

Separate complete native timing. The completed trained-head timing study evaluates all nine methods on 128 reused ordinary-panel roots, nine repeats and three seeds, using the final V3 heads. Its 31,104 timings include complete native input construction and scoring, with weights loaded outside the timed region. Joint pin processing takes a median 6.948 ms, versus 6.105 ms for WLDN. The median *paired* joint/WLDN ratio is 1.130502 (13.1% slower), rather than the quotient of those marginal medians. All 54 warmups, timed outputs and 3,456 separately replayed audit decisions have zero score error and no changed choices. These are descriptive shared-host measurements, not a cross-machine speed comparison or throughput estimate. The added cost does not rescue the failed quality criterion.

Reproduction and claim boundary. The saved quality protocol, numeric report and audit are retained under `chess-pin-quality-v3` in the project evidence directory; the timing study is under `chess-pin-trained-cost-v2`. Their audit receipts begin `c86562f02e45` and `80415131be41`. The separate manuscript table receipt binds full audit, figure and source hashes. The publication helper `write_pin_quality_v3.py` checks those identities before producing Tables 14 and 15. These results establish neither Elo nor gameplay strength, a general speed advantage, independent confirmation, biological superiority or a novel mechanism advantage.

H SUBMISSION STATUS

This file is a development draft targeting the ICLR 2027 format, not an accepted, submitted, or scientifically complete paper. The official call checked on September 18, 2026 lists September 18 at 23:59 AoE for abstract registration and September 25 at 23:59 AoE for the full paper. No registration is performed by building this document. Authorship, human review, a defensible final contribution, complete evidence and an anonymous supplementary package remain outstanding.